



Experiment No: 9

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Subject Name: System Design

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1- Aim - To design a scalable Online Movie Ticket Booking System

2- Requirements: Functional & Non-Functional

A- Functional Requirement

- User Registration & Authentication: Users can sign up, log in, and manage profiles using email/password or OAuth (Google, Facebook).
- Movie Browsing: Users can browse currently running movies, upcoming releases, with filters by genre, language, city, and rating.
- Theatre & Showtime Selection: Users can view all theatres in a city showing a selected movie along with available showtimes.
- Seat Selection: Users can interactively view the seat layout (Regular, Premium, Recliner) and select available seats in real-time.
- Booking & Payment: Users can book selected seats, proceed to payment via multiple gateways (UPI, Cards, Net Banking, Wallets), and receive booking confirmation.
- Booking History & Tickets: Users can view past and upcoming bookings, download e-tickets, and cancel bookings within the allowed window.
- Notifications: Users receive email and SMS notifications on successful bookings, cancellations, and upcoming show reminders.

B- Non-Functional Requirement

- 50–100 million users; 10K concurrent bookings/sec during peak.
- Availability: 99.99% uptime (< 1 hr downtime/year)
- Strong consistency for seat booking; Eventual for reviews
- < 200ms for search; < 500ms for seat lock
- Fault-tolerance: No single point of failure; auto-failover

3- Core-entities of System

- User
- Movie
- Theatre
- Seats
- Booking
- Payment
- Review

4- API endpoint creation



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1. User-Onboarding

POST /api/users/register
POST /api/users/login
PUT /api/users/{userID}/profile
GET /api/users/{userID}/bookings

2. Movie Browsing & Search

GET /api/movies
GET /api/movies/{movieID}
GET /api/movies/search
GET /api/movies/{movieID}/shows

3. Seat Selection & Booking

GET /api/shows/{showID}/seats
POST /api/shows/{showID}/lock
POST /api/bookings
GET /api/bookings/{bookingID}
DELETE /api/bookings/{bookingID}

4. Payment Service

POST /api/payments/initiate
POST /api/payments/webhook
GET /api/payments/{paymentID}

5- High-Level Design

